CSE321 Quiz 03 SET A
Time 30 Minutes

Q1.You are making a File System. You have a total of 500 data blocks, 3 inode blocks. Each block size is 4kb (2^12 byte). The memory address bits needed for each block is 128 byte (2^7 byte). Each Inode takes 256 kb(2^8) of memory inside the inode blocks. Knowing these information, answer the following questions

a. If Inode structure has 8 direct blocks, 1 single indirect block and 2 double indirect blocks. What's the max size of the file? [2.5]
b. The user wants to copy a file from an external source where there are a total of 58 files inside that folder, each file contains 1 block of data. Justify with math on whether there will be an issue to achieve this or not. [2.5]

Q2.You have a file named "Motivation Quotes" where there are 2 blocks of data written inside the file. You want to delete the file. Given that Open("cse321/Motivation Quotes") is already done for you. Simulate the access timeline flow [3]

	Data bitmap	Inode bitmap	Cse321 inode	Target file inode	CSE321 data	Target file Data [0]	Target file Data [1]

Q3. [2]

a. In the three criteria while writing a data: data bitmap update, inode update, data block write. In which point a crash has to appear for the situation to be labelled as "space leak"
b. If the OS crashes after a journal commit but before the file is actually written. Now OS restarts, what will the file system do to ensure consistency?